# Gobinath Kulathunga Ashwinthan

*Electrical and Electronic Engineering Graduate — RTL Design & Verification*

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## INTRODUCTION

Electrical and Electronic Engineering graduate focused on digital hardware design for FPGAs. Hands-on experience across communications systems, signal processing hardware, and automated design verification, taking designs from specification through simulation to working FPGA implementation. Seeking an entry-level role in RTL design and verification, with a strong interest in computer architecture and processor verification. Fluent in English, Tamil, and Sinhala.

## TECHNICAL SKILLS

**Programming**: MATLAB, Python, ARM Assembly, Verilog HDL, VHDL
**Libraries Used**: NumPy, SciPy, Pandas, Matplotlib, TensorFlow, PyTorch, OpenCV, Scikit-learn
**Microcontrollers/Boards**: ESP32, Arduino (AVR), Artix-7 (Basys-3), Zynq UltraScale+ MPSoC (XCZU9EG)
**Tools**: Arduino IDE, Proteus, KiCad, Vivado Xilinx, PS-CAD

## PROJECTS

### MIL-STD-188-110B HF Modulator — FPGA RTL Design & Implementation

*May - Jun 2026*

*VHDL, Vivado, MATLAB, AXI4-Stream, Avalon-ST, ILA, Zynq UltraScale+ ZCU102*

**Overview:** Designed and implemented a **MIL-STD-188-110B-compliant HF modem transmit chain** in synthesizable **VHDL**, supporting multiple data rates (**150–1200 bps**) and interleaving configurations.

**Contributions:** Developed key DSP and baseband modules including **CDC** interfaces, 1/2 rate convolutional **FEC encoder**, block **interleaver**, **known-data insertion**, and **8-PSK symbol mapper**, with a parameterized RTL architecture and efficient dataflow control. Performed MATLAB-based functional verification, RTL simulation, and timing optimization, and integrated the design on the **AMD Xilinx Zynq UltraScale+ ZCU102** at **500 MHz** using **AXI4-Stream** and **Avalon-ST** interfaces and on-chip logic analyzers (**ILA**) with JTAG.

### Fixed-Point Standard Deviation IP — FPGA DSP Architecture & Optimization

*Jul - Aug 2026*

*VHDL, MATLAB, Vivado, Xilinx CORDIC IP, BRAM LUTs, DSP Slices, Zynq UltraScale+*

**Overview:** Designed and implemented a **parameterized, synthesizable VHDL IP core** for **fixed-point standard deviation** computation, targeting FPGA-based digital signal processing applications. Built a resource-efficient **two-pass architecture** covering fixed-point addition, subtraction, multiplication, division, and **square-root computation using Xilinx CORDIC IP**, with optimized word lengths and numerical precision.

**Contributions:** Optimized the datapath using **BRAM-based reciprocal lookup tables**, **DSP slice utilization**, and **arithmetic restructuring**, achieving a **37.5% reduction in DSP usage** (**16 to 10 slices**). Performed **MATLAB-based fixed-point modeling**, RTL verification, and timing/resource optimization on the **AMD Xilinx Zynq UltraScale+** platform.

### Automated RTL Verification Framework — FPGA Demodulator

*Sep - Oct 2026*

*Python, Tcl, Verilog, VHDL, XSim, QuestaSim, Regression Testing*

**Overview:** Developed a **Python-based RTL verification framework** integrating **Verilog testbenches** and **Tcl scripting** for automated **system-level** and **incremental bottom-up module verification** of an FPGA demodulator.

**Contributions:** Implemented **dynamic RTL configuration** using **VHDL generate constructs**, enabling flexible module integration and targeted debugging through command-line arguments. Integrated **XSim** and **QuestaSim** with configurable batch/GUI execution, reducing simulation runtime through selective compilation and direct simulator invocation. Automated **test-vector generation**, **golden-reference comparison**, output logging, and **regression testing** for **bit-accurate functional verification**.

### 8-bit ALU Design

*Dec 2025*

*Verilog HDL, Digital Arithmetic Modules*

**Overview:** Designed a fully structural 8-bit Gate-Level ALU supporting **signed** and **unsigned** arithmetic with a **ripple-carry two's complement adder–subtractor** and explicit overflow detection and multiplication using a **sequential shift-and-add architecture** and division via an **iterative restoring division algorithm** with shared subtract–compare logic. Integrated multi-cycle control (**start/busy/done handshake**) and a complete flag set: zero, negative, overflow, carry-out, and divide-by-zero.

### Parameterized Synchronous FIFO in Verilog

*Feb 2026*

*Verilog HDL, RTL Verification, Memory Architecture*

**Overview:** Designed and verified a **parameterized single-clock synchronous FIFO** in Verilog RTL with configurable **depth** and **data width**. Implemented a **circular buffer** using phase-extended read and write pointers for reliable full and empty detection. Verified correct operation across **pointer wraparound** and **concurrent read/write conditions**. Developed a **task-based testbench** with directed scenarios and waveform analysis to validate data orderingand status flag behavior.

### Sobel Edge Detection using Verilog

*Feb 2026*

*Verilog HDL, FPGA, VGA Controller*

**Overview:** Implemented a real-time streaming Sobel edge detection pipeline on Artix-7 (Basys-3) using Verilog RTL, processing **640×480 VGA at 25 MHz**. Designed **dual-BRAM line buffers** and a **3×3 window generator** to achieve one-pixel-per-clock convolution. Integrated RGB565 image ROM, fixed-point grayscale conversion, pipelined gradient computation, saturation, and dynamic threshold control. Demonstrated multiple **display modes** (RGB, grayscale, edge, overlay) with precise VGA timing and latency alignment.

**Other Projects:** Subsurface emergency transmitter using near-field magnetic induction (KiCad PCB design); image compression using Huffman coding (Python).

## EDUCATION

**VLSI CAD: Logic (Part I) — Ongoing Online Course** — Ongoing
*Coursera, University of Illinois Urbana-Champaign*

**VHDL for FPGA Design** — 2026
*AMD (Xilinx) Adaptive Computing Training*

**Designing with Vivado Design Suite** — 2026
*AMD (Xilinx) Adaptive Computing Training*

**MATLAB Onramp** — 2026
*MathWorks*

**B.Sc. Engineering (Hons.) in Electrical and Electronic Engineering (GPA - 3.31)** — 2022 – 2026
*University of Peradeniya*

**G.C.E. Advanced Level and Ordinary Level Examinations** — 2012 – 2021
*Royal College, Colombo 07*

**Primary Education** — 2007 – 2011
*Colombo Hindu College, Bambalapitiya*

## EXTRACURRICULAR ACTIVITIES

**Karate - University of Peradeniya** — 2023
*Sri Lanka University Games (SLUG 2023)*

- Represented University of Peradeniya in Karate at the Sri Lanka University Games 2023, showcasing discipline, teamwork, and competitive spirit.

**Diploma in Electronic Keyboard** — 2025 – Present
*London College of Music (LCM), University of West London*

- Pursuing a diploma in Electronic Keyboard through the London College of Music, refining musical expression and proficiency.

Resume - soft copy

## REFERENCES

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